

Saliency, Ranking, and Metabolism: Three Conflated Signals in Long-Running Agent Memory Systems

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Abstract

Long-running agent memory systems commonly treat "importance" as a single quantity. We argue that it conflates three distinct signals, and that the conflation is a structural design error rather than an implementation detail. A scoring formula written in 2023 for a simulation demo—relevance plus recency plus importance—moved into production hearts without a checkup, letting the three jobs of "importance" (sedimentation, ranking, retirement—which we later separate as saliency, ranking, and metabolism) interfere with one another within a single signal. Our position rests on a service chain: memory serves the LLM; the LLM serves the human—and memory must be *memory*: self-describing and self-metabolizing, its saliency and retirement decided by data and lifecycle rather than one-shot human-delegated verdicts (Invariant Io). Skip the middle link and even the most elaborate memory system drifts toward RAG. From 8 months of production telemetry on a single conversational system (earliest data 2026-02; ~38K raw memory items, ~148K association links as of the 2026-08 audit, the live store now holding ~108K), we argue that "importance" is at least three orthogonal signals that are best kept physically separated: saliency (intrinsic to the data), ranking (a query-time service), and metabolism (lifecycle retirement). We document three production failures of conflation—a 93× semantic skew inside one counter, a structure-blind quality metric reporting 0.999 while the store's structure-aware coherence read 0.101, and 88% of raw memories being "islands" (73% at the time of writing, still declining without cleanup)—yet containing the most load-bearing roots. The retrieval-side pressure is measurable: an access-distribution Gini coefficient of 0.960, with the top 1% of nodes absorbing 51.9% of all accesses. We contribute a signal taxonomy with four invariants (Io–I3), six design laws, and two same-day audit tools (the Silence Test and the RAG Test). No benchmark supremacy is claimed; all evidence comes from our own failures, of which we have plenty.

Keywords: agent memory; signal separation; positive-feedback bias; RAG; memory governance; experience report

1. Introduction: For Whom Does Memory Serve—the Legacy of RAG

"Can thinking take effect next time?—But you've already forgotten what you sorted out last time."

—an utterance from our system's corpus (lightly redacted)

Before any scoring formula, a more fundamental question: for whom does memory exist?

Human memory is not built for queries: an event sinks, and years later surfaces unprompted—as a scent, a tone of voice, a habit of handling things. Humans forget by *living*—not by executing a deletion policy, but by being slowly overgrown with later life. Human memory has no index, and it never grades its own recollections.

An LLM starts from the opposite direction: by default every conversation begins fresh. Its memory cannot grow organically; it must be built.

Our position is a service chain: memory serves the LLM; the LLM serves the human. Remove the middle link—let memory serve the human's queries directly—and no matter how elaborately the system is dressed up (extraction, consolidation, graphs, layers), we contend it functions, in essence, as RAG: a person queries, a machine searches, the results are pasted into context with the model as glue. RAG as a tool is beyond reproach; as a memory architecture, we hold that it is a category error—it compresses the entire life of memory into the moment of retrieval, shaped by the query. Memory becomes an index; the model becomes a pipe.

Human memory and silicon memory are two different media; between media you need a bridge, not direct wiring. Pasting human documents raw into context is wiring; letting raw utterances settle into structure, letting derivations converge around anchors, keeping causal chains traceable—rebuilding the past in a way the model can grow on—that is a bridge. A line from the corpus says it better (lightly redacted): *"Don't design a bridge with an engineer's mind—feel the pull with physical intuition. Cracks close by themselves once they recognize each other; they are not built shut."* This is also our entire attitude toward human intervention: the structure of memory should be *recognized*, not manufactured. A bridge that can stand needs at least three different materials—our protagonists.

The service chain answers *for whom* memory serves; a second, equally fundamental question remains: *by what right* does memory hold together? Our secondary thesis: to give an LLM memory, the precondition is that memory be memory itself—not memory under human interference. Salience is not a score someone assigns; it is the data's own shape. Metabolism is not a deletion policy someone executes; it is memory living through its own lifecycle. The semantics of a database are binary—a record is either present or absent, and its verbs are insert, update, delete, query; the semantics of memory are alive—grow, converge, retire, circulate—all decided

endogenously by the data: data decides salience; data decides metabolism. Asking an LLM for a 1–10 score at write time, or an LLM judge for keep-or-delete verdicts, inserts a foreign verdict between the data and its own weight—the signature of human interference. A concurrent work [7] asks, "Is agent memory a database?" Our answer: no—and we offer the positive program of what it is: a self-consistent, metabolizing memory organism.

The current industry default is a retrieval-age one: it treats memory as something to be optimized for lookup. The concrete fossil of the retrieval mentality is a formula inherited from a demo. In 2023, a simulation of a town of 25 LLM agents (Generative Agents [1]) proposed what is now effectively the industry-default memory retrieval formula:

score = relevance (cosine similarity) + recency (exponential decay, factor 0.995, computed from last access) + importance (an LLM-assigned 1–10 score at write time)

The formula was written for a performance, yet became the skeleton of many production systems. The problem is not the summation—weighted fusion is innocent—but that it quietly lets one signal work three jobs: how *thick* a memory is (whether it is an anchor that later content converges around), where it *rank*s this time (whether it fits the present query), and how much longer it *lives* (cleanup logic widely reads counter/popularity-style inputs). Stuffing three signals into one field is like engraving heart rate, speed, and fuel consumption onto a single dial: the needle moves, but it no longer reports any one thing.

Our claim: in the system we studied, this is not roughness of implementation but a structural category error. Four contributions, each independently checkable:

- **C1 (Thesis)** A repositioning of the memory service chain: memory serves the LLM, the LLM serves the human; memory serving human queries directly tends to collapse into RAG.
- **C2 (Taxonomy)** A formal separation of the salience/ranking/metabolism signals, the memory-self-consistency principle, and four invariants (I0–I3).
- **C3 (Evidence)** Three production-grade failure cases (a 93× field-semantics skew, a 0.999 false-health metric, 88% island roots) with post-separation field data (93× → 2.2×; Gini 0.960).
- **C4 (Tools)** Two same-day executable audit tools: the Silence Test and the RAG Test.

Roadmap: §2 locates the lesion in related work → §3 gives the diagnostic framework → §4 submits the case files → §5–§6 derive design laws and audit tools → §7–§8 bound the claims and close. Finally, three readings to forestall. This paper is not against retrieval—retrieval is memory's legitimate act of access; we object only to its doubling as sole arbiter. It is not against scoring—a score may serve as input to the ranking layer, not as a proxy for salience or a termination criterion for metabolism. It submits no new architecture—what we deliver is a checkup framework and invariants that any existing system can self-test against without rebuilding.

2. Related Work: Three Heirs and a Critical Wave

(a) The textbook form—retrieval as reinforcement. In Generative Agents [1], recency decays from last access: a retrieved memory has its recency refreshed and thus becomes easier to retrieve again. This is the well-studied positive-feedback/popularity bias of recommender systems [5]; there its endpoint is called a filter bubble. Moved into a memory system, the endpoint is a memory bubble: a model left with only the things it likes to recall. To our knowledge, this specific critique has not been formalized for the agent-memory context.

(b) The fused form—one judge, three jobs. Memo [2] makes two LLM calls per write: extract candidate facts, then an LLM judge rules ADD / UPDATE / DELETE over vector-similar neighbors. Saliency judgment (how important is the new fact), metabolism decision (should the old fact die), and retention material (what survives for retrieval) are fused in a single ruling, triggered by the accident of "whoever similarity can retrieve." In fairness, the field is starting to feel the pain: the Memo team now discusses eviction and forgetting [3], and the community proposes access-frequency-driven lifecycle cleanup [4]. Where such work begins to separate retrieval decay from eviction, our distinct contribution is to generalize it into a full three-signal taxonomy with invariants (I0–I3) and to supply longitudinal production evidence for each failure mode; the problem is acknowledged, but a remedy is not yet formulated.

(c) The half-separated form—closest, one step short. Zep's [6] bi-temporal edges let superseded facts be invalidated rather than deleted (history preserved)—the most lucid handling of metabolism. But its retrieval ranking remains relevance + recency + episode usage—usage frequency enters ranking directly; behavioral contamination of the measure persists.

(d) The gathering critical wave. The concurrent work "Is Agent Memory a Database?" [7] criticizes current scoring as "local heuristics applied at retrieval time." We agree with the diagnosis and move two steps further: a complete three-signal taxonomy with invariants, and production-grade longitudinal failure evidence. [7] also proposes GEM, a framework whose correctness is defined at the state-trajectory level (four state-level operators—ingestion, revision, forgetting, retrieval—plus six correctness properties); our invariants instead define correctness at the signal level (fields and their writers). The two are complementary rather than competing: GEM governs what a memory operation must do to a state, whereas we govern which writer may touch which field. The window of criticism is opening; this paper tries to nail the frame.

(e) An anticipated rebuttal: why not just ask an LLM to score? The most natural defense: let an LLM score importance at write time ([1]'s 1–10), or judge at update time ([2])—doesn't that give us "importance"? Three reasons it fails. First, it violates I0: a one-shot external verdict substitutes for the data's own weight—and the judge cannot see the future; on write day no one knows which raw utterance will become the sole basis of a critical judgment three months later (§4.3). Second, the judge is retrieval's prisoner: an LLM judge only sees the

neighbors similarity can put in front of it, so "what counts as important" is hijacked by "what is similar" (§2b). Third, verdicts cannot metabolize: once written, a score becomes a lifetime label, and the system loses the ability for data to matter later—once written, the score becomes a permanent label—effectively a one-time in/out verdict.

Table 1: Three heirs × handling of the three signals

System	Salience (intrinsic)	Ranking (query service)	Metabolism (lifecycle)
Generative Agents [1]	none (LLM 1–10 at write, for life)	blended relevance + recency (access-refreshed) + importance	none
Memo [2]	fused with metabolism & retention in one LLM ruling	vector similarity + ruling outcome	DELETE inside the same ruling (triggered by similar neighbors)
Zep [6]	none	relevance + recency + usage (behavior in ranking)	bi-temporal invalidate-not-delete (closest to I2)
This paper	intrinsic to data, convergence measured in-scope, unaffected by behavior	bounded [0,100] + decay, never written back	item-level cascade, lifecycle-triggered, popularity-blind

3. Method: Signal Definitions, Invariants, and a Feedback Analysis

Notation: $\sigma(x)$ is the salience of item x , $\rho(x)$ its ranking-layer score, $\mu(x)$ its metabolic state; the three have distinct domains and distinct writers—this is the section's entire claim. On ρ : the stored score is an input to per-query ordering, fused at query time with query similarity; the persistent score itself carries no query.

Definition 1 (Salience). An intrinsic structural property of x : *how much derived content converges around x as anchor*, measured within the correct scope (counting across semantic domains yields fan-out noise, not convergence). Key property: unaffected by retrieval behavior. It is the shape of the data, not its popularity; it grows while no one retrieves. Concretely, our implementation measures it as a normalized convergence density: $\sigma(x) = \log_{1p}(\text{physical in-degree from anchor-forming edges}) / \log_{1p}(p_{99})$, where p_{99} is the 99th-percentile in-degree over those same anchor edges, counted only over anchor edge types (raw_basis, ai_basis, derived_from)—recall-driven edges are excluded by construction, so σ stays flat under retrieval.

Definition 2 (Ranking). The output of a query-time algorithmic service; a bounded counter (e.g., [0,100]) with forgetting-curve decay is a legal implementation. Key properties: recomputable at will; serves only the

consumer; never written back to data. Our implementation: an Ebbinghaus-style score $\rho = \text{count_norm} \times \exp(-\Delta t / S)$, where count is the recall count (hard-capped at 100), count_norm is that count normalized to [0,1] by the cap, Δt is days since last recall, and $S = 7 \times (1 + \text{count})$ carries the same day units as Δt .

Definition 3 (Metabolism). A lifecycle decision, cascading at the item level (when a raw utterance retires, its derivations and links retire with it). Key property: triggered by lifecycle events, not by any popularity or ranking score. Low frequency $\neq$ expired—coldness is the birthright of root nodes, not their crime. Our implementation: retirement is a soft status mark with item-level cascade; it reads lifecycle events only, never a counter. Figure 1 summarizes the three domains and the walls that keep them apart.

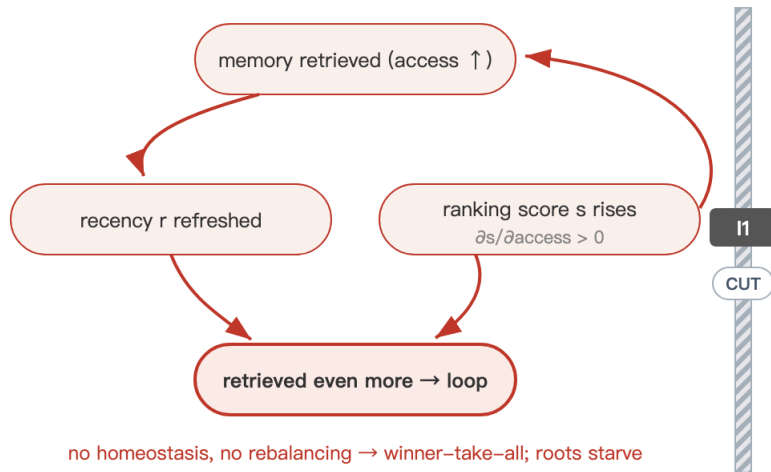
A deeper footnote to the separation. Another corpus line (lightly redacted): *"The essence of time is interaction and experience—time itself is just a virtual unit of measurement."* Ranking's recency decay treats time as a unit of measurement; nothing wrong there—the clock is the ranking layer's legitimate tool. But salience is not a function of the clock; it is sediment of interaction—constituted by how much derivation converges, regardless of the wall clock. Time belongs to the clock; interaction belongs to memory. This is the seam along which ranking and salience must part.

Invariants:

- **I0 (No proxy):** memory's persistent properties (salience, survival) must not be assigned by one-shot external verdicts (LLM scores, LLM judge rulings)—they are endogenous to data and lifecycle;
- **I1 (No contamination):** no change in ranking signals may alter salience or the data itself;
- **I2 (No wrongful death):** metabolism must not be triggered solely by ranking/popularity scores;
- **I3 (No polysemy):** one physical field must not carry multiple semantics.

Feedback sketch (why conflation necessarily worsens) (Figure 3). Let ranking score s depend on recency r ; r decays over time but refreshes on access; then $\partial s / \partial \text{access} > 0$, while access probability rises with s —a self-reinforcing loop. Only by locking "access" strictly inside the ranking layer and forbidding any write-back to persistent fields (I1) is the loop physically cut. This is not engineering vanity; it is the minimal structural requirement for cutting a feedback loop. The magnitude of the pressure is measurable: in this system's current (post-separation, bounded-and-capped) ranking layer—25,705 nodes, an access-count Gini coefficient of 0.960 (Figure 7), the top 1% of nodes absorbing 51.9% of all accesses, the top 10% taking 96.9%, and 85.8% of nodes with zero accesses. The correct reading: winner-take-all in the ranking layer is legal—it is confined to the service layer, recomputable at will. The point of the number is the warning: a distribution like this, written into any persistent field, is a retirement sentence for 85.8% of zero-access nodes—and most of them are roots.

Conflation: self-reinforcing loop (memory bubble)



Separation: one-way service; loop gone

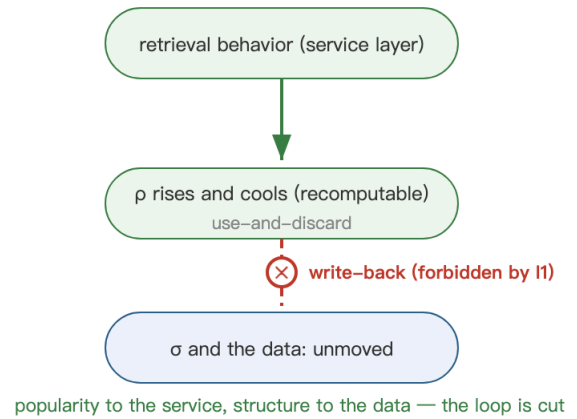
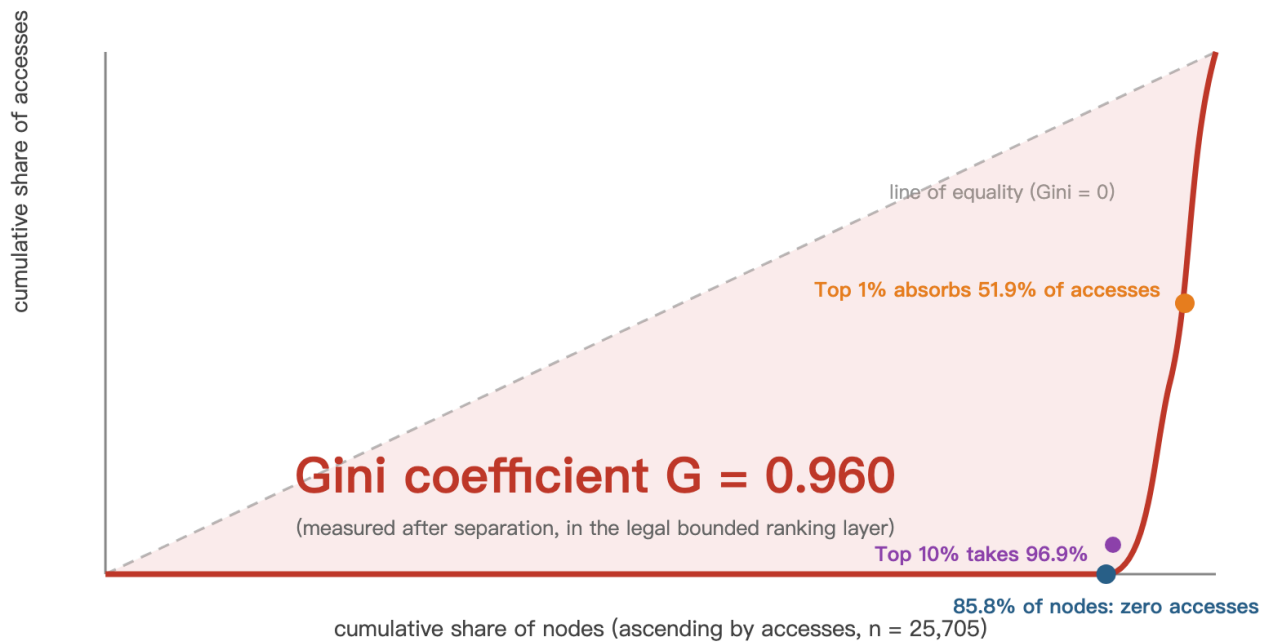


Figure 3 | The positive-feedback loop and how I1 cuts it. Left: under conflation, retrieval refreshes recency (Generative Agents' 0.995 decay is access-based), $\partial s / \partial \text{access} > 0$ forms a self-reinforcing loop; without homeostasis the endgame is winner-take-all. Right: I1 locks access sensitivity inside the service layer and forbids write-back; the loop is physically cut. Human contrast: the testing effect's reinforcement is bounded by synaptic homeostasis and rebalanced in sleep consolidation — imitate all of it.



Written into any persistent field, this distribution is a retirement sentence for the 85.8% zero-access nodes (most of them roots) — this is why I1 exists

Figure 7 | Lorenz curve of the ranking-layer access distribution (production-measured, 2026-08). 25,705 nodes: Gini 0.960; the top 1% of nodes take 51.9% of all accesses; the top 10% take 96.9%; 85.8% have zero accesses. Measured after the three-signal separation — winner-take-all in the ranking layer is legal (bounded, recomputable, never written back); the figure quantifies the pressure and explains why no persistent field may reference such a distribution (Invariant I1).

The life of one memory (a worked example) (Figure 2). Put the three definitions on one timeline. Day 0: a raw utterance enters; in the wired world it receives an LLM score of 5 and lives with that lifetime label; in the bridged world it has no score, only an identity—raw anchor. Day 30: it happens to be retrieved twice; in the wired world its recency refreshes, its score rises, and it begins to attract more retrieval; in the bridged world its ranking rises too—but all of it stays inside the service layer; its salience is untouched, still zero convergence. Day 90: no one asks; in the wired world its score decays to zero and it enters the cleanup queue; in the bridged world it is equally "cold"—but only its ranking is cold, not its right to exist. Day 120: a new judgment needs a basis, and finds this is the only anchor; in the wired world it was deleted by popularity on day 100—the judgment loses its basis while the dashboard glows green; in the bridged world it is surrounded by derivation, and its salience grows for the first time—from island to root. One memory, two fates; the difference lies not in any single algorithmic step, but in whether the three jobs are separated.

The human-memory contrast (and one honest boundary). Human retrieval does strengthen memory—the testing effect is a classic result of cognitive psychology [8]; is I1's "no access write-back" anti-biological? No—biology is being quoted halfway: human strengthening occurs in a separate synaptic substrate, bounded by homeostatic renormalization, and rebalanced wholesale during sleep consolidation [9]—reinforcement with a ceiling, a ledger, and a rebalancing. Software counters have only reinforcement, no homeostasis: the unbounded positive feedback is exactly the memory bubble of §2a. I1's correct reading is not "don't imitate biology" but "imitate all of it": access sensitivity may live in the ranking layer (bounded, decaying, recomputable); the persistent layer must remain behavior-independent—unless a separate mechanism with homeostasis and rebalancing is introduced.

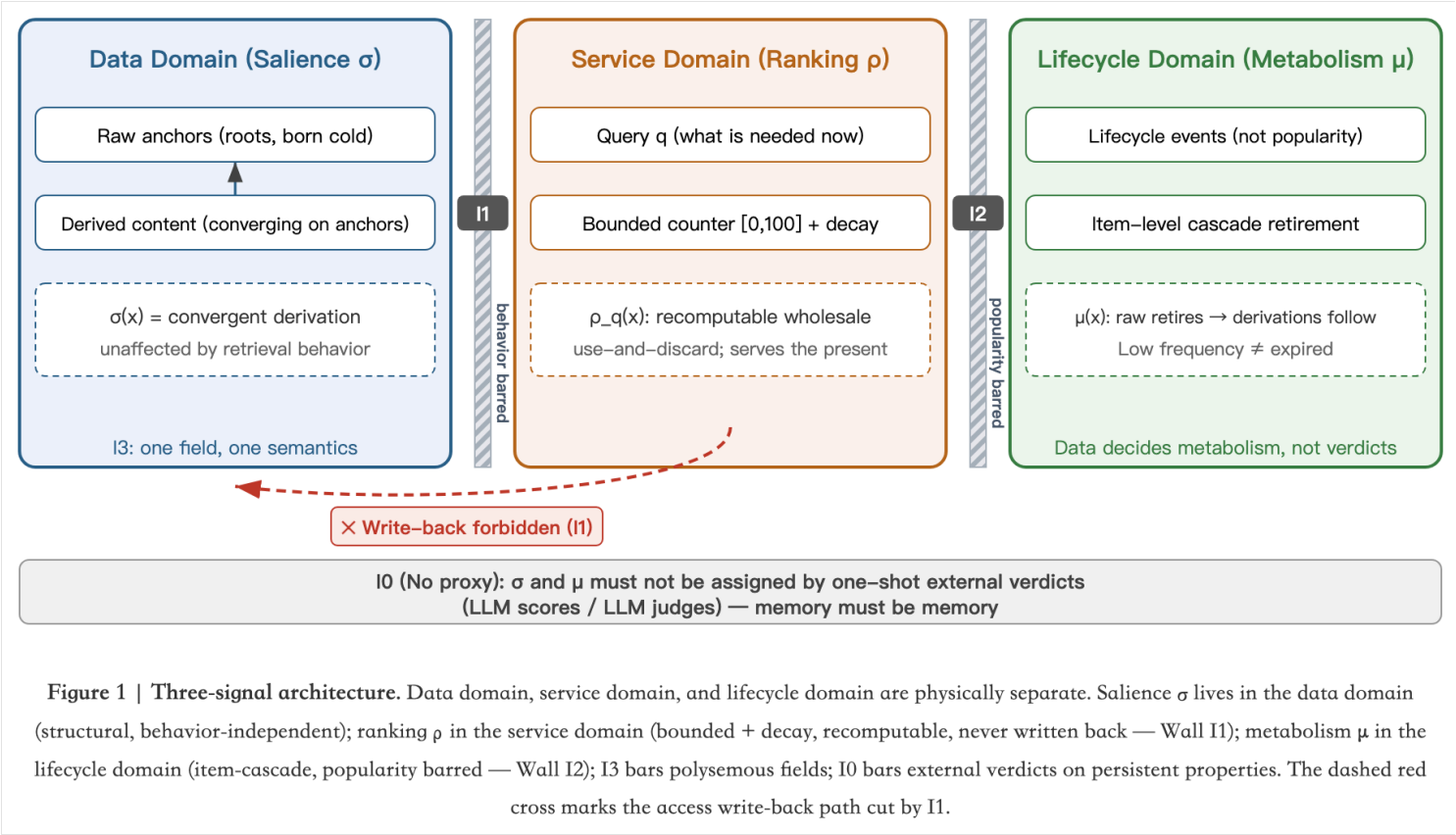


Figure 1 | Three-signal architecture. Data domain, service domain, and lifecycle domain are physically separate. Saliency σ lives in the data domain (structural, behavior-independent); ranking ρ in the service domain (bounded + decay, recomputable, never written back — Wall I1); metabolism μ in the lifecycle domain (item-cascade, popularity barred — Wall I2); I3 bars polysemous fields; I0 bars external verdicts on persistent properties. The dashed red cross marks the access write-back path cut by I1.

4. Case Study: Production Failure Evidence (single system, longitudinal, n=1)

System background: a conversational agent memory system running continuously for ~8 months, dual-store (raw utterance layer + distillation layer), ~38,000 memory items, ~148,000 association links (2026-08 audit snapshot; the live store now holds ~108K). All numbers below come from real replays of the production store.

Timeline of the evidence (single store):

time point	what happened	cohort
early audit (pre-fix)	full-store audit finds the 93× field-semantics skew and 88% island share	whole raw layer at audit
post-fix (field split, cap [0,100])	cross-type span 93× $\rightarrow$ 2.2×; peak pinned at 100	same field, re-measured
2026-08 (fixed snapshot)	island share 67.8% (user, n=5,956) / 82.6% (AI, n=3,173); combined 73.0% (n=9,129); Gini 0.960 (n=25,705)	raw utterances that month

time point	what happened	cohort
2026-10 (live store)	~108K links; raw layer ~13,364 (user 6,289 + AI 7,075)	current

4.1 First page: one field, three semantics, a silent 93× distortion

In the early design, one count field was written by three paths: retrieval +1, distillation +1, another subsystem +1. No one noticed—until an early full-store audit produced the X-ray: "principle"-type items averaged 449, "raw utterance"-type items averaged 4.8. Same field, same decay curve, a 93× span across semantics; a single peak of 1965 became a black-hole item absorbing all ranking attention. Every ranking and cleanup routine reading that field was doing arithmetic on three incommensurable quantities daily, while the reports stayed perfectly normal.

The fix was not tuning—tuning cannot save a field shared by three semantics—it was splitting: one field per semantics, one decay constant each, uniformly capped at [0,100]. Post-separation field data (measured 2026-08): the cross-type span compressed from 93× to 2.2× (principle 2.9 vs raw 1.3—the residual gap is legitimate under unified semantics); the peak pinned at 100 (Figure 4).

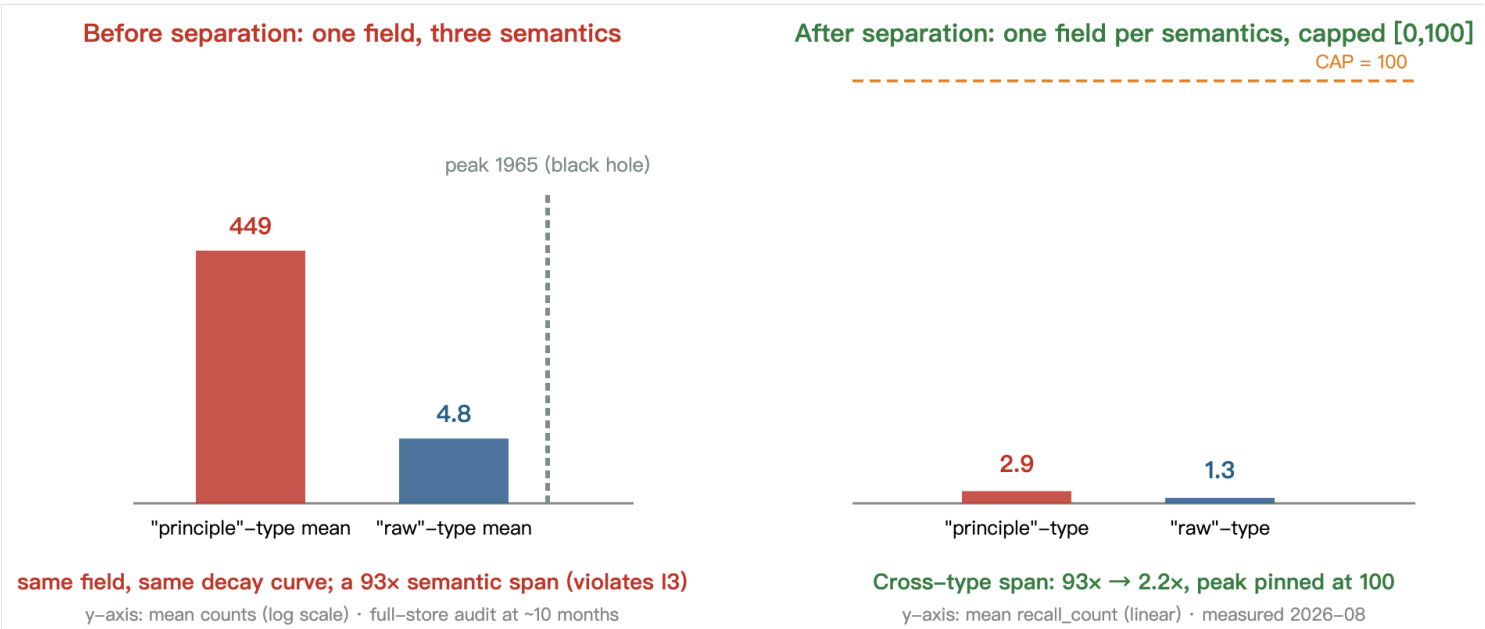
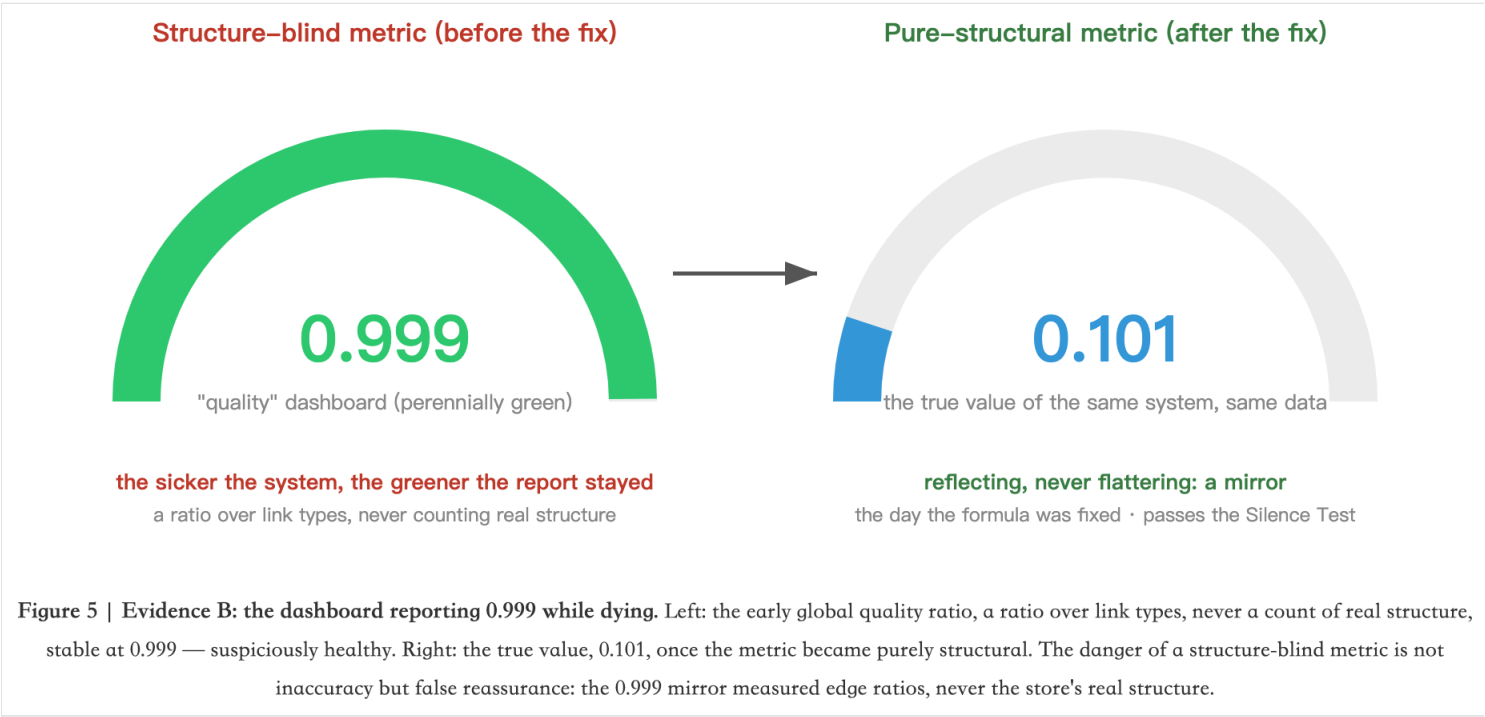


Figure 4 | Evidence A: counter semantics distortion and the separation fix (production-measured). Left (log scale): before separation, one count field written by three paths — a 93× span between type means (449 vs 4.8), a single peak of 1965 forming a black hole (violates I3). Right (linear): after splitting fields and capping at [0,100], the cross-type span is 2.2× (legitimate under unified semantics); the peak is pinned.

4.2 Second page: a dashboard reporting 0.999 while dying

An early global ratio assessed memory quality, stable at 0.999 for ages—suspiciously healthy. We once took it for a grade. Root cause: the metric was a ratio over link types, not a count of real structure—so it never tracked the store's health. Concretely, the early metric was coherence = (total_links - contradict) / total_links; because contradict edges are rare across the whole store, the ratio pinned near 1 (0.9996, displayed as 0.999) and read green while the store degraded. The corrected metric counts true inter-standing-wave reinforcement density: coherence = log1p(d) / log1p(3), where d = 2 × (unique reinforce pairs) / (standing-wave node count), computed over the standing-wave pool alone. The standing-wave pool is the set of live cortex nodes that carry a completed vector and a weight above the activity gate; nodes below that gate stay in the raw layer and are not counted here. On the same store it read 0.101.

The most dangerous property of a structure-blind metric is not inaccuracy but the sicker the system, the greener the report stayed. A green dashboard does not mean the car is fine; it may only mean the dashboard is broken. A corpus line serves as this page's epigraph (lightly redacted): *"Isn't that just looking in a mirror?"* Figure 5 shows the before/after. A metric should be a mirror—reflecting, never flattering; the 0.999 mirror reflected the lens, not the road.



4.3 Third page: load-bearing memories can score zero under popularity metrics

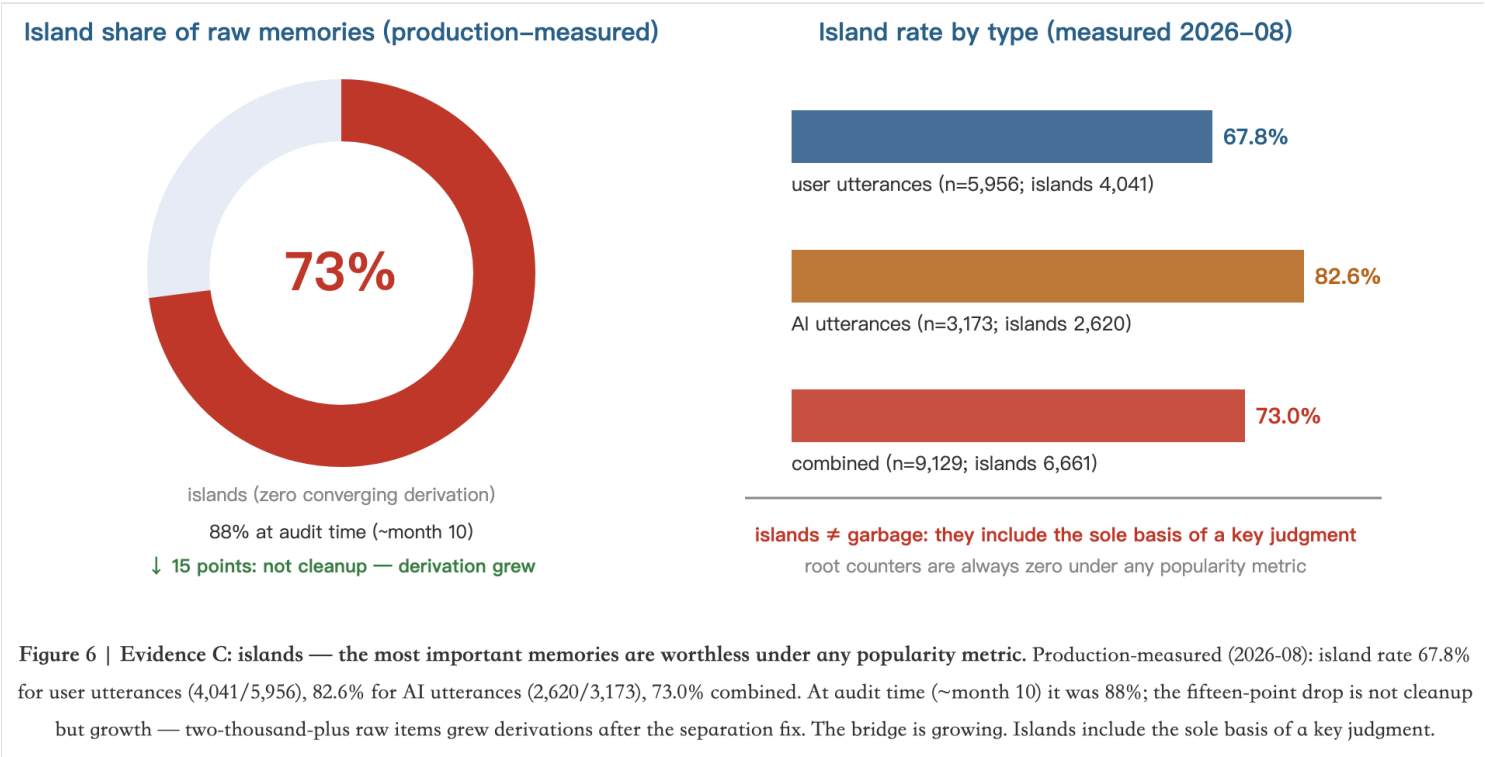
A full structural audit: 88% of raw items are islands—zero converging derivation. Operationally, we count an island as a raw utterance with $\sigma = 0$: no anchor-forming in-edges. The island share is a direct count over the

raw-utterance cohort (user 67.8%, n = 5,956; AI 82.6%, n = 3,173; measured 2026-08). They are not garbage; they are roots. Two real examples (lightly redacted):

First, a self-description of decision style: *"His way of pricing risk: not by computing the worst case, but by looking for the minimum gain."* A dozen words, no derivation around them, zero on every counter. Its future value does not depend on being retrieved—it is the bedrock of how a person judges; some future decision will rest on it, invisible to any popularity metric.

Second, a passing remark about the roof of an old family house—unvisited for three months, zero derivation, zero counts—until a judgment of later consequence was formed and it turned out to be the sole basis. For those three months, retrieval popularity rated it as having no value.

This is the whole case for I2: root nodes are born cold; that is structure, not anomaly. A popularity-triggered cleanup can delete a root that later becomes a key judgment's sole basis—and after deletion, the dashboard gets even greener. The post-fix follow-up gives this page its coda: at that early audit the island share was 88%; by this writing it has fallen to 73% (user utterances 67.8%, AI utterances 82.6%; measured 2026-08, n=9,129). The fifteen-point drop is not the merit of cleanup but the merit of being left alone: roughly fourteen hundred raw items (a 15-point drop on a cohort of about 9,100) have grown derivations with no one intervening. The bridge is not only on the blueprint; it is growing (Figure 6).



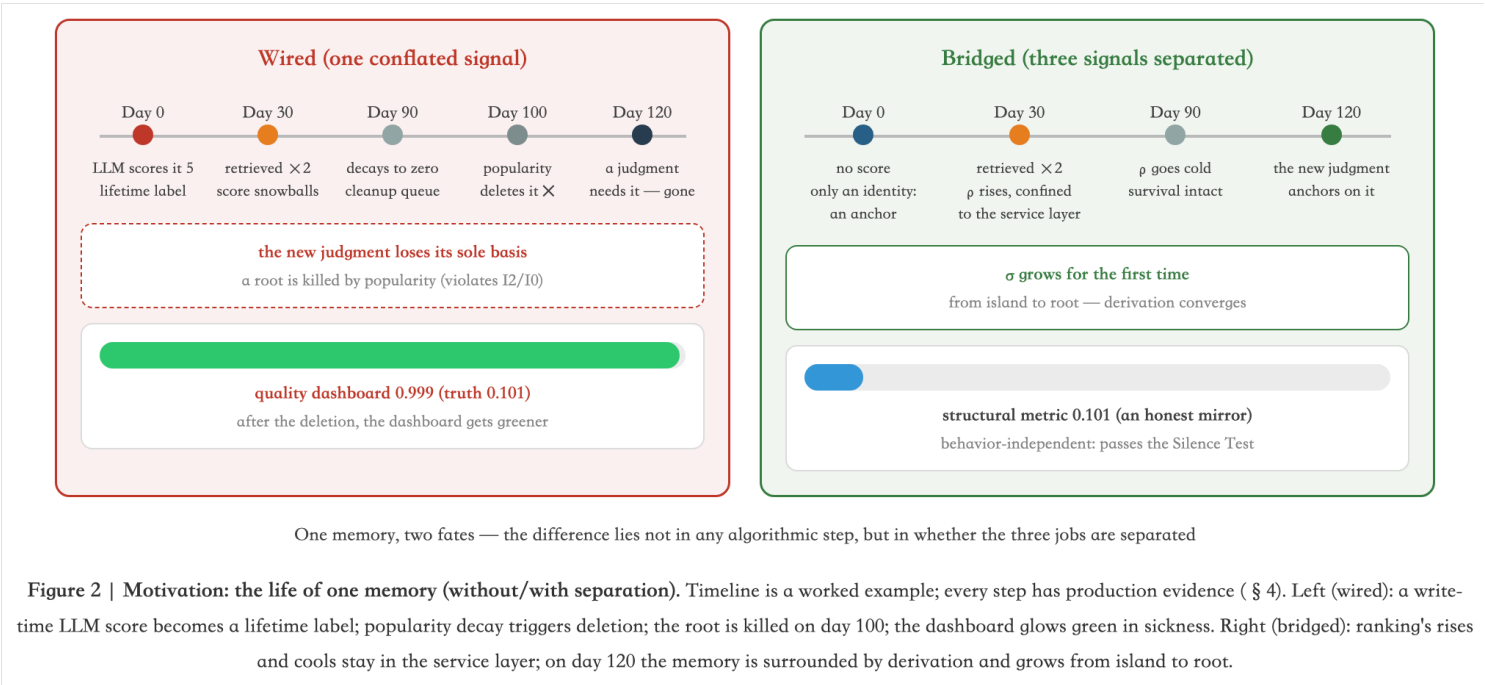


Table 2: Case overview—symptom, hard evidence, violated invariant, fix, post-fix field data

Case	Symptom	Hard evidence (measured)	Violates	Fix	Post-fix
§4.1	one field, three semantics	93× cross-type mean skew (449 vs 4.8); peak 1965 black hole	I3	split fields, per-semantics decay, cap [0,100]	93× → 2.2×; peak pinned at 100
§4.2	health reported in sickness	reported 0.999; structure-aware coherence 0.101	none — passes all of I0–I3	count real structure (inter-standing-wave reinforcement density)	now tracks the store
§4.3	popularity-driven deletion	audit 88% islands (now 73%), incl. the sole basis of a key judgment	I2 (write-time 1–10 scoring is an I0 violation)	lifecycle-triggered cascade metabolism	88% → 73%; ~1,400 items still deriving

Three pages, three seemingly unrelated symptoms—one common cause: human interference—the counter was a shortcut someone designed; the structure-blind formula was a gauge someone trusted; the popularity cleanup was a policy someone approved. In the three cases we examine, the cause was not a bug in the usual sense, but a well-meaning, convenient intervention.

5. Design Laws

1. **Salience is data, ranking is a service, metabolism is a lifecycle**—three signals, three mechanisms, physically separate fields.
2. **Counters never write back**: algorithmic processes may be recomputed from scratch; data they have altered cannot be restored.
3. **Measure salience in the correct scope**: counting edges across semantic domains counts fan-out.
4. **Metabolism cascades at the item level**, triggered by lifecycle; low frequency is not expiry.
5. **Polysemous fields are time bombs**—split early; every day late is another day of dirty values.
6. **Every importance metric must pass the Silence Test.**

6. Tools: Two Audits You Can Run Today

The Silence Test. Suspend all retrieval behavior for T days (or replay history offline while intercepting all writes to persistent counters), then recompute your "importance/quality/salience" metrics. If the numbers do not move, you measured data; if they drift, you measured popularity—a health report contaminated by behavior. Cost: T days of patience, nearly zero.

The RAG Test (a blunter cut). Replace your entire "memory layer" with a good-enough search engine. Fix, in advance, a workload and an outcome criterion; if behavior shows no material difference on that pre-declared criterion, the layer was, by that measure, retrieval only—not memory. For memory to be memory, there must be evidence beyond the query: salience that grows while unretrieved, metabolism that happens while unobserved, memories that shape the model's behavior without being asked about.

Silence-Test Day: a five-step protocol (half a day).

1. Snapshot all current "importance/quality/salience" metric values;
2. Freeze all retrieval paths for T days—or replay history offline, intercepting every write to persistent counters;
3. Recompute the same metrics; compare item by item;
4. Every drift item is a behavior-contaminated item; mark its upstream write path (usually the $\partial s / \partial \text{access}$ $\rightarrow$ feedback source);
5. Publish the result, good or bad. A system passing I1 should show zero drift. One honest limit: a zero result isolates access-driven contamination only; contamination arising from other update paths must be controlled separately before independence is claimed. We followed this boundary up empirically on our own store: across the ~38k-node cortex, salience σ and the access counter are weakly but non-trivially

correlated (Pearson 0.24), not zero. That residual is not a write path—it is a shared structural cause, since structurally central nodes are also more retrievable, and the correlation varies widely by node type (−0.17 to +0.50). The decisive evidence remains the formula: no access counter enters how σ is computed, and I1 constrains exactly that write path, not correlation.

We suggest maintainers schedule a Silence-Test Day and report results under this framework, favorable or not—what this field lacks is not new mechanisms but a first checkup of the old ones.

7. Discussion: Limitations, Validity Threats, and Why No Benchmark

Single system, single-user corpus, longitudinal; no cross-system benchmark (compute cost); the salience measure depends on the system's association-edge structure and needs adaptation to migrate; the evidence comes from failure retrospectives and may carry survivor-narrative bias—we judge the direction unaffected, but state it plainly; the corpus is dominated by private conversation, so cross-domain generalization is unverified; "the life of one memory" is a worked example whose every step has corresponding production evidence (§4) but whose timeline is synthetic.

Why no cross-system benchmark? Three honest reasons. Compute budget does not permit reproducing six classes of baselines. Mainstream benchmarks (e.g., LOCOMO) measure question-answering accuracy, while our diagnostic target is signal structure—a system can score well on the benchmark while violating all four invariants (the 0.999 dashboard of §4.2 is exactly such "high-scoring sickness"). And the future work we propose is precisely to metricize the Silence Test itself: "the drift of the importance metric after T days of suspended retrieval" as a comparable cross-system memory-health metric (drift $\approx$ 0 passes)—which requires no training compute at all, only honesty.

Data and artifact availability. All aggregate statistics (count distributions, island shares, before/after contrasts) are archived as anonymized distribution plots in the public preprint record (<https://doi.org/10.5281/zenodo.23068930>); the raw corpus will never be released for privacy; the five-step Silence-Test protocol is a public artifact, freely reusable and citable without permission.

8. Conclusion

Finally, the state of the system after the fix. Post-separation: cross-type distortion converged from $93\times$ to $2.2\times$, counter peaks pinned at 100; the quality metric became purely structural, counting real structure rather than edge ratios; the 88%-island raw utterances are alive to this day, many still deriving. Worth emphasizing: these

three fixes introduced no new "clever algorithm"—no new model, no new scorer, no learned weights—only the three conflated responsibilities separated, so that salience and metabolism are decided by the data once more. That is all.

What we would rather leave with the reader is a less comfortable judgment: as long as the industry keeps intervening in LLM memory by hand—scoring in place of sedimentation, verdicts in place of metabolism, wiring human queries directly into memory past the middle medium—the cumulative effect is not better memory but a system that degrades under continued hand-tuning. RAG at least identifies itself as a retrieval engine; presenting one as a memory system while removing everything that happens beyond the query is a subtler and more misleading substitution.

And if you have ever written a boost, set a threshold, or tuned an LLM-judge prompt: maintainers who have written such boosts, thresholds, or judge prompts are, in this sense, the executors of that degradation. The good news: the checkup sheet is in §6—five steps, half a day, and you can start today.

The three signals are now stored separately in our system. We invite maintainers to examine their own: many may find that the metric has been tracking retrieval behavior rather than memory all along.

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